Gourav Singh

Bengaluru, Karnataka
+91-7477221021 | singhgopal08579@gmail.com | LinkedIn | GitHub

Education

Bangalore Institute of Technology, Bengaluru **2023 – 2027**
B.E. in Robotics and Artificial Intelligence Engineering — CGPA: 8.45/10

Kendriya Vidyalaya No.2 BSF, Indore **2022**
CBSE Class XII: 92.6% — Class X: 92.6%

Technical Skills

Languages: C, C++, Python, JavaScript
Frameworks: ReactJS, Node.js, ExpressJS, REST APIs
Databases/Cloud: MongoDB, PostgreSQL, Redis, Firebase
AI/ML: Machine Learning, TensorFlow, Scikit-learn, Gemini API
Core Concepts: DBMS, OOPs, Operating Systems, Networking Fundamentals, System Design
Tools: Docker, Git, GitHub, Linux, Postman, Prometheus

Relevant Coursework

Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Embedded Systems, Signals and Systems

Projects

MolGenix – AI-Powered Drug Discovery Pipeline | *ReactJS, FastAPI, PostgreSQL, Redis, TensorFlow* | **GitHub**

- Developed an end-to-end AI-driven computational drug discovery pipeline integrating NLP, generative molecular modelling, ADMET safety prediction, and molecular docking workflows.
- Implemented FastAPI orchestration backend with PostgreSQL and Redis to coordinate molecule generation, chemical validation, and real-time processing modules.
- Integrated RDKit, DeepChem, and AutoDock Vina for molecular validation, toxicity screening, and physics-based binding affinity analysis of generated drug candidates.

ChaosLab – Chaos Engineering & Auto-Repair Platform | *Node.js, React, Docker, Prometheus, Redis, k6* | **GitHub**

- Built an end-to-end chaos engineering platform that performs automated fault injection (CPU spikes, memory exhaustion, network latency, container kills) against containerised target applications under simulated traffic load using k6.
- Engineered an auto-repair engine that monitors real-time Prometheus and cAdvisor metrics, detects threshold violations, and autonomously applies remediation strategies including container scaling, rollback, and rate limiting.
- Designed a Node.js control plane with BullMQ job queues, Toxiproxy network fault proxying, Dockerode container orchestration, and a React dashboard with live Socket.io log streaming and Recharts metric visualisation.

TechCart E-Commerce Platform | *ReactJS, Node.js, ExpressJS, MongoDB* | **GitHub**

- Developed a scalable full-stack e-commerce application supporting product browsing, cart, and checkout workflows.
- Designed RESTful backend APIs using Node.js and ExpressJS with MongoDB integration for efficient data handling.
- Built responsive frontend interfaces using ReactJS to improve usability and application performance.

Certifications

• Stanford University Machine Learning Coursework • Google Generative AI Certification • Postman API Fundamentals Student Expert • TLE Eliminators Data Structures and Algorithms

Achievements

- Qualified for Smart India Hackathon Round 2 with an AI-based student dropout warning system.
- Winner, Project Expo 2025 at BIT for CNN-based AI project implementation.
- Secured 2nd place at HackSurgeX Hackathon for developing an AI-driven drug discovery solution.
- NDA 151 Course AIR-483 conducted by UPSC.

Extracurricular Activities

- Active member of Aero Club; participated in RC aircraft construction workshops and club outreach activities.
- Participated in national-level sports competitions and actively engaged in public speaking, debates, and quizzes.